8. PAEDIATRIC CANCER PAIN

According to The Malaysian Society of Paediatric Haematology and Oncology, the incidence of paediatric cancer in Malaysia is about 77.4 per million in children aged <15 years old.[111] The paediatric cancer pain is quite different from the pain in adults and children respond differently to treatment.

Pain is a common symptom in children diagnosed with cancer. The pain can be tumour-related, procedure-related or treatment-related.

Tumour-related pain can present:[112]
- before or at diagnosis
- during initial treatment
- when tumour is resistant to treatment
- at disease recurrence

The following table shows examples of cancer-related pain in children.

Table 6. Examples of Cancer-related Pain in Children:[112]

Tumour-related pain	• before or at diagnosis • during initial treatment • when tumour is resistant to treatment • at disease recurrence
Procedural related pain	a. diagnostic procedures • venepuncture • lumbar puncture • bone marrow aspirate and biopsy • tissue biopsy b. Procedures • central venous line insertion • pleural or peritoneal drainage • external ventricular drainage • ventricular-peritoneal shunt • surgeries • wound dressing/debridement
Treatment-related pain	• mucositis (post-chemotherapy or radiotherapy) • acute pancreatitis (SE of chemotherapy e.g. asparaginase) • neutropenic enterocolitis • haemorrhagic cystitis (e.g. with cyclophosphamide, ifosfamide, radiotherapy) • intracranial haemorrhage (thrombocytopenia from bone marrow suppression) • peripheral neuropathic pain (e.g. with vincristine, cisplatin) • post-operative pain • phantom limb pain • procedural pain (on treatment protocol)

8.1 Principles of Pain Assessment

Pain assessment in children can be quite different than adults. The following are principles to guide pain assessment in children using the acronym **A.B.C.D.E**:

Table 7. ABC of Pain Assessment in Children[112]

A	**A**sk the child and **A**ssess pain score
B	Use **B**ehavioural and Biological measures
C	Find the **C**ause
D	**D**ecide and **D**eliver treatment in a timely manner
E	**E**valuate outcome

Assessment and management of pain in children and infants are different from adults due to:[112]

- Communication: Children have limited verbal and cognitive abilities. Non-verbal cues and observation are essential approaches to assessment.
- Development: Children have a wide physiological, cognitive, and developmental response to pain according to their age groups. This has to be carefully assessed.
- Fear and anxiety: Children are reluctant to report pain that reflect their pain experience. Managing emotions can be done through play therapy, distraction and other cognitive behaviour strategies.
- Dosage: Medication must be carefully adjusted according to age, weight and developmental understanding. Adolescent also has a unique approach to cancer pain management. Healthcare workers will require appropriate training for assessment and management.
- Parental involvement: Caregivers input is an essential component and part of the assessment.
- Ethical consideration: Children are not able to give consent for any intervention and medical management. Consent must be taken from legal caregivers or parents.

8.2 Pain Assessment Tools

The choice of a pain assessment tool should take into consideration:[112, 113]

- the child's developmental age
- verbal ability

These are shown in **Table 8** and **9**.

Table 8. Pain Assessment Tools Based on Child's Developmental Age[112]

Age	Pain rating scale
1 month to 4 years	**FLACC** • Observe the child's behaviour in 5 dimensions (Face, Legs, Arms, Cry, Consolability) for 2 to 5 minutes, and assign a score (maximum 10)
4 years to 7 years	**Revised FACES** • Picture-based scale where the child selects 1 to 6 faces to represent their pain experience
≥7 years	**Numerical rating scale** • Ask the child to assign a number to their pain, with '0' being no pain and '10' being the worst imaginable pain

Table 9. Pain Assessment Tools Based on Verbal Ability[112]

Special population	Pain rating scale
Neurological impaired	**Revised FLACC** • Incorporates individualised pain behaviours which is unique to a child
Critically ill	**COMFORT-Behaviour scale and FLACC**
Neonates	**Neonatal/Infant Pain Scale (NIPS)**

8.3 Treatment

Cancer pain in children can be effectively managed by using drugs e.g. opioids, non-opioids and adjuvant analgesics with the biopsychosocial or multi-modality approach covering physical, psychosocial and spiritual entities.

- WHO uses simple principle for analgesia in children:[112]
    - oral route is the preferred choice
    - dosing of analgesic should be at a fixed regular interval
    - WHO 3-step analgesic ladder is the proposed model

Analgesia is given based on severity of pain from mild to severe pain in the 3-step WHO ladder in children. Weak opioids still have a role despite insufficient robust data.[112]

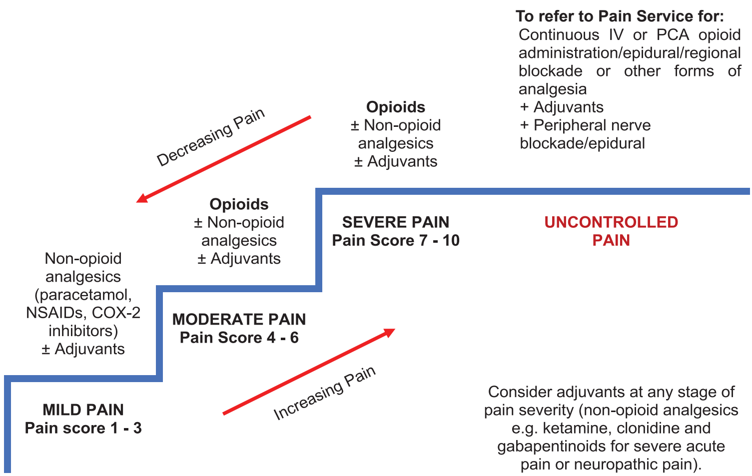

Figure 3. Modified Analgesic Ladder

Source: Ministry of Health, Malaysia. Paediatric Pain Management Guidelines 2023. Putrajaya: MoH; 2023.

In developing countries, children with cancer experience pain related to advanced disease and treatment approaches e.g. chemotherapy or radiotherapy.[114]

- **Non-opioid analgesics**

The first step in the WHO Analgesic 3-Step Ladder is non-opioid drugs, e.g. paracetamol and NSAIDs, with the optional use of adjuvants for mild pain.[112] In the second step of WHO pain ladder, non-opioid analgesics may work synergistically with opioids as co-analgesic to produce a better pain relief (refer to Figure 3).[112]

Paracetamol is generally safe but may cause hepatotoxicity if overdosed (refer to **Appendix 5b**).[115] GI ulceration, nephrotoxicity and CV events are the known AEs of NSAIDs. Thus, lowest effective dose of NSAIDs should be given with proton pump inhibitors to prevent gastro-duodenal ulcers.[116] The dosages of commonly use non-opioid analgesics are shown in **Appendix 5b**.

Recommendation 17
- Paracetamol or nonsteroidal anti-inflammatory drugs should be used in children with mild cancer pain.
- Paracetamol should be used in combination with opioids as co-analgesic unless contraindicated in children with cancer pain.

- **Opioid analgesics**

Morphine is considered in the second step of WHO 3-Step Ladder when the pain is moderate to severe in children. The minimum interval between each dose is between one to four hours. The benefits of using an effective strong opioid analgesic outweigh the benefits of weak opioids in the paediatric population when compared with the uncertainty associated with the response to codeine and tramadol. Caution on the use codeine and tramadol has been issued due to ultra metabolisers and potential AEs.[117, level III]

a. Tramadol

Tramadol is a synthetic analgesic with unpredictable effects due to its wide variability in metabolism. The drug has a ceiling effect. It is unsuitable for escalating mild-moderate pain or severe pain. It has the potential to cause side effects in children.[112]

b. Morphine

Oral morphine is the first-line therapy for severe cancer pain in children. Its effectiveness in pain relief has been extrapolated from the treatment of adult with chronic cancer pain.[112] Oral morphine is available as either IR or SR preparations. IV morphine is used for rapid onset analgesia and when the patients are unable to tolerate oral morphine.[117, level III] If the opioid requirement goes beyond 1 mg/kg/day, it is likely that the patient will require regular morphine.[113; 115] Morphine dose should be monitored after 24 - 48 hours of morphine use. Alternative routes of administration should be based on clinical judgement, drug availability and patient's preference. The initial dose of morphine and its frequency is shown in **Appendix 5b**.

Case Example for Opioid Titration in Paediatric Cancer Pain

Aiman is a 10-year-old boy with relapsed Acute Lymphoblastic Leukemia with bone metastasis. He complains of generalised pain with pain score of 6/10. He is opioid naïve with normal renal and liver function. His weight is 20kg and he is currently at home.

Method of dose calculation:

- Immediate release oral morphine 0.2 mg/kg/dose x 20 kg = 4 mg q4h (maximum initial dose is 5 mg/dose for children) = 24 mg/day

Breakthrough dose: 1/10 to 1/6 of daily dose (2.5 - 4 mg), can be served 1-2 hours after previous dose of morphine.

Recommendation 18

- Oral morphine is the preferred choice for children with moderate to severe cancer pain.

c. Fentanyl

Transdermal fentanyl is an effective alternative to oral morphine in children with difficulty in swallowing or those having intractable nausea and vomiting whose opioid requirements are stable.[115] IV fentanyl can be used in children with specific circumstances e.g. renal failure but preferably this is done under specialist care.[112] Refer to **Table 4** on suggested dose conversion ratio for conversion of oral morphine to fentanyl patch.

d. Oxycodone

Oxycodone is an alternative strong opioid which is as effective as oral morphine. Refer to **Table 4** on suggested dose conversion ratio for conversion of oral morphine to oxycodone.

Recommendation 19
- Fentanyl or oxycodone may be used as alternative analgesics in children with moderate to severe cancer pain.

e. Methadone

Methadone is only used as an alternative opioid for cancer pain in children. However, it should only be prescribed under specialist supervision in palliative care settings.[118, level III]
For opioid AEs and their management, refer to **Chapter 4.4.8**.

- **Adjuvant drugs**

Adjuvant analgesics may be used with other analgesics including strong opioids in children with cancer pain.[11] Combining drugs with different mechanisms of action improve analgesia and decrease AEs in the patients. This can be used at any stage of pain severity as per **Figure 3**. However, there is insufficient evidence on the use of adjuvant analgesics in the paediatric age group.

The use of antidepressants in children has been associated with an increased risk of suicidal ideation and behaviour.[119] However, amitriptyline has been used in the management of pain especially bone pain and neuropathic pain in children.[120, level III]

In children, neuropathic pain can be treated with anticonvulsants e.g. gabapentin, pregabalin and sodium valproate. It is important to monitor undesired AEs e.g. headache, drowsiness and ataxia when commencing these agents.[113]

Ketamine should be used by specialists familiar with cancer pain management in children. It is generally used in low doses.[115]

Corticosteroids are commonly used for children with pain related to mass effect of tumour e.g. headache from brain metastases, abdominal pain from liver capsule distension or intestinal obstruction and neuropathic pain from spinal cord compression.[121]

Bisphosphonates should be considered where analgesics and/or radiotherapy are inadequate for the management of painful bone metastases.[120, level III; 121; 122, level III]

Refer to **Appendix 5b** on Dosage of Commonly Used Adjuvant Drugs in Children with Cancer Pain.

Recommendation 20

- Adjuvant analgesics may be considered in cancer pain management in children.

9. BARRIERS AND EDUCATION

- **Barriers**

Barriers to the effective management of cancer pain need to be identified and addressed. A systematic review found negative attitudes and a lack of knowledge towards cancer pain management among the healthcare providers, patients, family caregivers and general public. The most commonly cited barriers were fear of drug addiction, tolerance and AEs of opioids.[123, level III]

In another study on cancer pain management by family caregivers, the main challenges can be grouped into three parts:[124, level III]

- communication and teamwork issues which included caregivers' receipt of inadequate information regarding pain management and, inappropriate and ineffective communication from the healthcare team
- caregiver issues which were related to caregivers' fear and beliefs, concurrent responsibilities and, lack of pain-related knowledge and skills
- patient issues which included patient's own fear and beliefs, psychological and physiological well-being, adherence to medications and reluctance to report pain

A cross-sectional study conducted in Hospital Umum Sarawak, Malaysia showed that:[125, level III]

- among the four domains of patient-related barriers explored via BQ-II questionnaire, fatalism had the highest median BQ-II score, followed by harmful effects, physiological effects and communication
- education level, gender and marital status had significant impact on various barrier domains
- ethnicity had no significant impact on all four domains

A multinational cross-sectional survey showed:[126, level III]

- of all the attitudinal barriers, fear of addiction to opioids was the strongest barrier across all countries whereas fatalism was the weakest barrier
- barriers scores were higher in patients of older age, male gender, higher pain severity or pain interference, lower Karnofsky scores and shorter duration of opioid use

In a multicentre cross-sectional study, depression was significantly associated with total barrier score to cancer pain management. Therefore, screening and treatment of depression should be an important component of successful pain management.[127, level III]

- It is important to identify and address barriers to effective cancer pain management.

- **Education**

Education on issues related to cancer pain is an essential element to effective cancer pain management. This involves not only the patients, but also the carers, family and healthcare providers.

Educational strategies should focus on addressing the following issues:[9]
- understanding cancer pain
- understanding disease processes and their relation to pain
- how to describe and document pain assessment appropriately
- understanding pain management
- awareness of the available analgesics
- dispelling fears regarding opioid analgesia
- accessing help and support (when, where and who)

10. FOLLOW-UP AND REFERRAL

Follow-up care for patients with cancer pain can be provided at home, primary care clinics or specialised outpatient clinics. With the advent of better internet services, teleconferencing or video call services can also be used to help patients who do not have easy access to conventional follow-up.

Two recent observational studies supported the structured outpatient follow-up of cancer patients:

- proper clinic guideline programme with a multidisciplinary approach, availability of pain interventions and palliative care referral in a specialist outpatient clinic led to significant improvement in BPI and pain score in ESAS[128, level II-3]
- physician-pharmacist joint clinic was significantly more effective than standard care in BPI pain intensity, adequacy of pain management and medication adherence[129, level II-2]

A home care service provided by community palliative care providers can reach out to patients in their own homes. This is especially important for patients who are unable to travel or have mobility issues. A Cochrane systematic review showed mixed results in the improvement in pain control between community home palliative care services and standard care.[130, level I]

In the previous local CPG on cancer pain, regular follow-up either at home, primary care clinics or specialised outpatient clinics including palliative care and cancer pain clinics according to their preferences or circumstances has been recommended.[9]

There are many different types of healthcare technology that can be used in delivering patient care. Videoconferencing can help when in-person conversations are not feasible. A Cochrane systemic review on telephone interventions for adults with cancer showed limited and mixed results on pain reduction compared with usual care. The certainty of the evidence on this outcome was very low.[131, level I] In a non-randomised controlled trial, home telemonitoring significantly increased pain registration and prescription for analgesics compared with usual care in cancer patients.[132, level II-1]

In another systematic review assessing the effectiveness of mHealth applications (apps) for self-management in improving pain, psychological distress, fatigue or sleep outcomes in adult cancer survivors, three out of four studies reported improvement in pain but only one showed a significant difference in those patients using the apps compared with those not using it.[133, level I]

A meta-analysis on the effectiveness of telemedicine on pain management in patients with cancer showed it improved:[134, level I]

- pain intensity (SMD= -0.28, 95% CI -0.49 to -0.06)
- pain interference (SMD= -0.41, 95% CI -0.54 to -0.28)

According to the Cochrane Risk of Bias, the risk of bias in most studies was considered low.

A list of community palliative care providers available in Malaysia can be downloaded from the Malaysian Hospice & Palliative Care Council website (https://www.malaysianhospicecouncil.com/).

Recommendation 21

- Cancer patients should be followed-up for pain management either in the specialist outpatient clinic, primary care clinic or at home.
  - Teleconsultations and digital applications may be used for this purpose.

Although most pain experienced by the patients can be managed by the primary team, there might be pain which does not respond well to initial treatment and requires specialised care. Thus, patients with severe pain and inadequate pain management should be considered for referral to pain or palliative specialist services.[135; 136, level III]

11. IMPLEMENTING THE GUIDELINES

Implementation of this CPG is important as it helps in providing quality healthcare services based on the best and most recent available evidence applied to local scenario and expertise. Various factors and resource implications should be considered for the success of the uptake in the CPG recommendations.

11.1 Facilitating and Limiting Factors

Existing facilitators for application of the recommendations in this CPG include:

- availability of the CPG to healthcare providers (hardcopies and softcopies/online)
- regular seminars/conferences/courses for healthcare providers on management of cancer pain including those involving professional bodies
- presence of "Pain as 5th Vital Sign" strategy and pain-free hospital concept involving multidisciplinary team
- public awareness activities that involve governmental/non-governmental organisations e.g. World Hospice and Palliative Care Day

Limiting factors in the CPG implementation include:

- limited awareness and understanding/knowledge in managing cancer pain among healthcare providers especially on the use of opioids
- variation in clinical management and preferences
- insufficient resources in terms of expertise and medications
- misconception about the disease and its management by the public

11.2 Potential Resource Implications

Appropriate assessment and treatment of cancer pain is crucial for the successful management of the condition. This includes the proper use of opioids for moderate to severe cancer pain. However, barriers to its use dampen the aims of controlling cancer pain at various level of care. These are supported by local studies on the issue and the lower rate of opioid use in the country compared with the global rate. Efforts must be improved to educate both healthcare providers and patients/carers on the judicious use of opioids in patients with cancer pain. Working together with the NGOs like hospice will facilitate the implementation of these guidelines. Certain medications like IR morphine should be made available in all primary care centres.

In line with the key recommendations of the CPG and the National Key Performance Index of Palliative Medicine, the following is proposed as clinical audit indicator for quality management of cancer pain:

$$\text{Percentage of patients with cancer pain score of 7 - 10 who are prescribed with strong opioids} = \frac{\text{Number of patients with cancer pain score of 7 - 10 who are prescribed with strong opioids in a period}}{\text{Total number of patients with cancer pain score of 7 - 10 in the same period}} \times 100\%$$

Implementation strategies will be developed following the approval of the CPG by MoH which include launching of the CPG, Quick Reference and Training Module.

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Appendix 1

EXAMPLE OF SEARCH STRATEGY

Clinical Question: What are the effective and safe pharmacological treatments in cancer pain?

1. CANCER PAIN/
2. ((cancer or neoplasm or tumo?r) adj2 associated pain).tw.
3. ((cancer or neoplasm or tumo?r) adj2 related pain).tw.
4. ((cancer-associated or cancer-related or cancer or neoplasm-associated or neoplasm-related or tumo?r-associated or tumo?r-related) adj1 pain*).tw.
5. 1 or 2 or 3 or 4
6. (pharmaco* adj1 treatment*).tw.
7. DRUG THERAPY/
8. (drug adj1 therap*).tw.
9. pharmacotherap*.tw.
10. ANALGESICS, NON-NARCOTIC/
11. ((non-opioid or nonopioid) adj1 (analgesic* or drug*)).tw.
12. (non opioid adj2 (analgesic* or drug*)).tw.
13. ((nonnarcotic or non-narcotic) adj1 (analgesic* or drug*)).tw.
14. ACETAMINOPHEN/
15. acet#minophen.tw.
16. n-acetyl-p-aminophenol.tw.
17. paracetamol.tw.
18. ANTI-INFLAMMATORY AGENTS, NON-STEROIDAL/
19. ((non-steroidal or nonsteroidal) adj2 (anti-inflammatory agent* or antiinflammatory agent*)).tw.
20. (nonsteroidal anti inflammatory adj3 agent*).tw.
21. (analgesic* adj1 (antiiflammatory or anti-inflammatory)).tw.
22. (non steroidal anti inflammatory adj4 agent*).tw.
23. (analgesic* adj2 anti flammatory).tw.
24. nsaid*.tw.
25. TRAMADOL/
26. tramadol.tw.
27. tramadol hydrochloride.tw.
28. TAPENTADOL/
29. tapentadol.tw.
30. tapentadol hydrochloride.tw.
31. CODEINE/
32. codeine.tw.
33. codeine phosphate.tw.
34. dihydrocodeine.tw.
35. ANALGESICS, OPIOID/

36. ((full or partial) adj2 opioid agonist*).tw.
37. (opioid adj1 analgesic*).tw.
38. (opioid mixed adj2 agonist-antagonist*).tw.
39. opioid mixed agonist antagonist*.tw.
40. opioid*.tw.
41. MORPHINE/
42. (morphine adj1 chloride).tw.
43. (ms adj1 contin).tw.
44. duramorph.tw.
45. morphine.tw.
46. (morphine adj1 (sulfate or sulphate)).tw.
47. (oramorph adj1 sr).tw.
48. OXYCODONE/
49. oxycodone.tw.
50. (oxycodone adj1 hydrochloride).tw.
51. oxycodone naloxone.tw.
52. FENTANYL/
53. fentanyl.tw.
54. (fentanyl adj1 citrate).tw.
55. atypical opioid.tw.
56. BUPRENORPHINE/
57. buprenorphine.tw.
58. (buprenorphine adj1 hydrochloride).tw.
59. NALBUPHINE/
60. nalbuphine.tw.
61. (nalbuphine adj1 hydrochloride).tw.
62. MEPERIDINE/
63. meperidine.tw.
64. (meperidine adj1 hydrochloride).tw.
65. pethidine.tw.
66. METHADONE/
67. methadone.tw.
68. (methadone adj1 hydrochloride).tw.
69. 6 or 7 or 8 or 9 or 10 or 11 or 12 or 13 or 14 or 15 or 16 or 17 or 18 or 19 or 20 or 21 or 22 or 23 or 24 or 25 or 26 or 27 or 28 or 29 or 30 or 31 or 32 or 33 or 34 or 35 or 36 or 37 or 38 or 39 or 40 or 41 or 42 or 43 or 44 or 45 or 46 or 47 or 48 or 49 or 50 or 51 or 52 or 53 or 54 or 55 or 56 or 57 or 58 or 59 or 60 or 61 or 62 or 63 or 64 or 65 or 66 or 67 or 68
70. 5 and 69
71. limit 70 to (english language and humans and yr=”2010-Current”)

Appendix 2

CLINICAL QUESTIONS

1. What are the principles of management of pain in patients with cancer?
2. What are the accuracy and reliability of clinical assessment tools of patients with cancer pain?
3. What are the accuracy and reliability of neuropathic pain assessment and tools in patients with cancer?
4. What are the accuracy and reliability of screening tools for comprehensive assessment of cancer pain?
5. What are the accuracy and reliability of pain assessment tools in patients with cognitive impairment/learning disabilities with cancer pain?
6. What are the principles of pharmacological treatment in cancer pain?
7. What are the effective and safe pharmacological treatments in cancer pain?
8. Are cannabinoids/medical cannabis effective and safe for treatment of cancer pain?
9. What are the prescribing, titration and maintenance issues of morphine and other strong opioids in patients with cancer?
10. What are the clinical issues related to tolerance, dependence, and addiction to opioids in patients with cancer?
11. What are the pharmacological strategies for breakthrough pain and other acute pain crises in patients with cancer?
12. What are the effective and safe adjuvant medications in cancer pain management?
13. What are the effectiveness and safety of different drug formulations and routes of administration in managing pain for patients with cancer?
14. What are the effectiveness and safety of anticancer therapies in the management of cancer pain?
15. What are the effectiveness and safety of radionuclide therapies in the management of cancer pain?
16. What are the effective and safe non-pharmacological/non-invasive treatments in cancer pain?
17. What are the effectiveness and safety of neurolytic therapies in management of cancer pain?
18. What are the effectiveness and safety of intrathecal neuraxial opioid and/or neuraxial adjuvants in refractory cancer?

19. What are the effectiveness and safety of surgery in cancer pain?
20. What are the effective and safe treatment for cancer pain in children?
21. What are the roles of multidisciplinary team/members/clinic in managing patients with cancer pain?
22. How should patients with cancer pain be followed-up?
23. What are the referral criteria of patients with cancer pain to be referred to specialist care in primary/secondary/tertiary care?

Appendix 3

SEDATION SCORE (MACINTYRE)

Score	Sedation level	Clinical findings
0	None	Patient is awake and alert
1	Mild	Occasionally drowsy, easy to rouse, and can stay awake once awoken
2	Moderate	Constantly drowsy, still easy to rouse, unable to stay awake once awoken
3	Severe	Somnolent, difficult to rouse, severe respiratory depression

Source: Macintyre PE & Schug SA. Acute Pain Management: A Practical Guide. Saunders Elsevier: London; 2007.

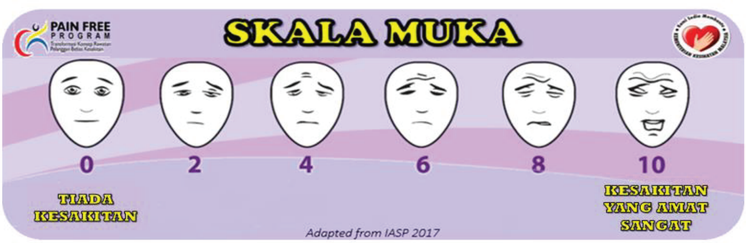
Appendix 4

ASSESSMENT TOOLS

a. Ministry of Health (MoH) Pain Scale

The MOH pain scale is a scale that combines NRS, the VAS and faces scale. The patient is asked to indicate his/ her level of pain intensity by pointing along a scale. The scale has numbers and the pain score is recorded as a number from 0 to 10.

In children less than 7 years old and cognitively impaired adults, other scales like IASP Faces Pain Scale or FLACC scale can be used. In patients who are sedated and intubated, pain assessment will rely on observations and behavioral assessment.



Source: Ministry of Health, Malaysia. Pain as the 5th Vital Sign Guideline: 3rd Edition. Putrajaya: MoH; 2018.

b. FLACC Scale

This is an observational score, and is used for paediatric patients aged >1 month to 3 years. It may also be used in adult patients who are unable to communicate verbally, e.g. very elderly patients, cognitively impaired patients.

1. Observe behaviour
2. Select a score according to behaviour
3. Add the scores for the total

Each of the five categories (F) face, (L) legs, (A) activity, (C) cry and (C) consolability is scored from 0-2, resulting in the total range of 0-10.

Category	Scoring		
	0	1	2
Face	No particular expression or smile	Occasional grimace or frown, withdrawn, disinterested	Frequent to constant quivering chin, clenched jaw
Legs	Normal position or relaxed	Uneasy, restless, tense	Kicking or legs drawn up
Activity	Lying quietly, normal position, moves easily	Squirming, shifting back and forth, tense	Arched, rigid or jerking
Cry	No cry (awake or asleep)	Moans or whimpers; occasional complaint	Crying steadily, screams or sobs, frequent complaints
Consolability	Content, relaxed	Reassured by occasional touching, hugging or being talked to, distractible	Difficult to console

Source: Ministry of Health, Malaysia. Pain as the 5th Vital Sign Guideline: 3rd Edition. Putrajaya: MoH; 2018.

c. Verbal Rating Scale (VRS)

No pain	**0**
Mild pain	**1**
Moderate pain	**2**
Severe pain	**3**
Very severe pain	**4**

The VRS consists of a list of adjectives describing different levels of pain intensity. Patients are asked to select the adjective that best represents their pain. This should reflect the extremes of this dimension; from ‘no pain’ to ‘very severe pain’ and sufficient intervening adjectives to capture gradations of pain intensity that may be experienced between extremes. VRSs are scored as above but these are ranks, not equal intervals.

Adapted: Outcome measures. The Faculty of Pain Medicine of the Royal College of Anaesthetists. 2019.

d. Short-form McGill Pain Questionnaire (SF-MPQ-2)

Date: _______________
Subject ID: ___________

For this questionnaire, I will provide you a list of words that describe some of the different qualities of pain and related symptoms. Please rate the intensity of each of the pain and related symptoms you felt during the past week on 0 to 10 scale, with 0 being no pain and 10 being the worst pain you can imagine. Use 0 if the word does not describe your pain or related symptoms. Limit yourself to a description of the pain related to your surgery or pelvic pain.

1.	**Throbbing pain**	*none*	0	1	2	3	4	5	6	7	8	9	10	*worst possible*
2.	**Shooting pain**	*none*	0	1	2	3	4	5	6	7	8	9	10	*worst possible*
3.	**Stabbing pain**	*none*	0	1	2	3	4	5	6	7	8	9	10	*worst possible*
4.	**Sharp pain**	*none*	0	1	2	3	4	5	6	7	8	9	10	*worst possible*
5.	**Cramping pain**	*none*	0	1	2	3	4	5	6	7	8	9	10	*worst possible*
6.	**Gnawing pain**	*none*	0	1	2	3	4	5	6	7	8	9	10	*worst possible*
7.	**Hot-burning pain**	*none*	0	1	2	3	4	5	6	7	8	9	10	*worst possible*
8.	**Aching pain**	*none*	0	1	2	3	4	5	6	7	8	9	10	*worst possible*
9.	**Heavy pain**	*none*	0	1	2	3	4	5	6	7	8	9	10	*worst possible*
10.	**Tender**	*none*	0	1	2	3	4	5	6	7	8	9	10	*worst possible*
11.	**Splitting pain**	*none*	0	1	2	3	4	5	6	7	8	9	10	*worst possible*
12.	**Tiring-exhausting**	*none*	0	1	2	3	4	5	6	7	8	9	10	*worst possible*
13.	**Sickening**	*none*	0	1	2	3	4	5	6	7	8	9	10	*worst possible*
14.	**Fearful**	*none*	0	1	2	3	4	5	6	7	8	9	10	*worst possible*
15.	**Punishing-cruel**	*none*	0	1	2	3	4	5	6	7	8	9	10	*worst possible*
16.	**Electric-shock pain**	*none*	0	1	2	3	4	5	6	7	8	9	10	*worst possible*
17.	**Cold-freezing pain**	*none*	0	1	2	3	4	5	6	7	8	9	10	*worst possible*
18.	**Piercing**	*none*	0	1	2	3	4	5	6	7	8	9	10	*worst possible*
19.	**Pain caused by light touch**	*none*	0	1	2	3	4	5	6	7	8	9	10	*worst possible*
20.	**Itching**	*none*	0	1	2	3	4	5	6	7	8	9	10	*worst possible*
21.	**Tingling or ‘pins and needles’**	*none*	0	1	2	3	4	5	6	7	8	9	10	*worst possible*
22.	**Numbness**	*none*	0	1	2	3	4	5	6	7	8	9	10	*worst possible*

23. Present Pain Intensity (PPI) - Numerical Pain Rating Scale. On a scale from zero to ten, zero indicating no pain and ten indicating worst pain imaginable, rate your pelvic pain:

none	0	1	2	3	4	5	6	7	8	9	10	*worst possible*

24. Evaluate overall intensity of total pain experience. Please check (✓) the word that describes the pain in your pelvic area only.

☐ *No pain* ☐ *Mild* ☐ *Discomforting* ☐ *Distressing* ☐ *Horrible* ☐ *Excruciating*

The SF-MPQ-2 consists of 24 different descriptors of pain of which each item is rated on a scale of 0-10. The scale of 0 equals to no pain and the scale of 10 equals to the worst ever pain experience during last week. The total score is calculated by summing all 24 scores.

Adapted: Melzack, Ronal. The short-form McGill pain questionnaire. Pain. 1987: 30 (2) :191-197.

e. Leeds Assessment of Neuropathic Symptoms and Signs (LANSS) Scale

Name________________________________ **Date**_________________________

This pain scale can help to determine whether the nerves that are carrying your pain signals are working normally or not. It is important to find this out in case different treatments are needed to control your pain.

A. PAIN QUESTIONNAIRE

- Think about how your pain has felt over the last week.
- Please say whether any of the descriptions match your pain exactly.

1) Does your pain feel like strange, unpleasant sensation in your skin? Works like pricking, tingling, pins and needles might describe these sensations.

a) NO – my pain doesn't really feel like this……………………………………………(0)
b) YES – I get these sensations quite a lot………………………………………………(5)

2) Does your pain make the skin in the painful area look different from normal? Words like motthled or looking more red or pink might describe the appearance.

a) NO – My pain doesn't affect the colour of my skin……………………...…………….(0)
b) YES – I've noticed that the pain does make my skin look different from normal…..(5)

3) Does your pain make the affected skin abnormally sensitive to touch? Getting unpleasant sensations when lightly stroking the skin, or getting pain when wearing tight clothes might describe the abnormal sensitivity.

a) NO – My pain doesn't make my skin abnormally sensitive in that area…………….(0)
b) YES – I've noticed that the pain does make my skin look different from normal…..(3)

4) Does your pain come on suddenly and in bursts for no apparent reason when you're still. Words like electric shocks, jumping and bursting describe the sensations.

a) NO – My pain doesn't really fell like this……………………………………….…….(0)
b) YES – I get these sensations quite a lot………………………………………………..(2)

5) Does your pain feel as if the skin temperature in the painful area has changed abnormally? Words like hot and burning describe these sensations.

a) NO – I don't really get these sensations…………………………………..…………….(0)
b) YES – I get these sensations quite a lot………………………………………………..(1)

B. SENSORY TESTING

Skin sensitivity can be examined by comparing the painful area with a contralateral or adjacent non-painful area for the presence of allodynia and an altered pin-prick threshold (PPT).

1) ALLODYNIA

Examine the response to lightly stroking cotton wool across the non-painful area and then the painful area. If normal sensations are experienced in the non-painful site, but pain or unpleasant sensations (tingling, nausea) are experienced in the painful area when stroking, allodynia is present.

a) NO, normal sensation in both areas……………………………………...…………….(0)
b) YES, allodynia in painful area only……………………………………………………..(5)

2) ALTERED PIN-PRICK THRESHOLD

Determine the pin-prick threshold by comparing the response to a 23 gauge (blue) needle mounted inside a 2 ml syringe barrel placed gently on to the skin in a non-painful and then painful areas.

If a sharp pin-prink is felt in the non-painful area, but a different sensation is experienced in the painful area e.g. none/blunt only (raised PPT) or a vary painful sensation (lowered PPT), an altered PPT is present.

If a pin-prick is not felt in either area, mount the syringe onto the needle to increase the weight and repeat.

a) NO, equal sensation in both areas…………….…………………………………………(0)
b) YES, altered PPT in painful area…………….………………………………………..(3)

SCORING:

Add values in parentheses for sensory description and examination finding to obtain overall score.

TOTAL SCORE (maximum 24)…………………………………………………………………

If score <12, neuropathic mechanisms are unlikely to be contributing to the patient's pain
If score >12, neuropathic mechanisms are likely to be contributing to the patient's pain

The LANSS identifies patients with neuropathic pain by combining the score of a patient's verbal description of pain and the results of neurological examination. This tool has two parts-a patient completed section and a brief physical assessment. Five questions in the patient-completed section (maximum score 16) identify those who are experiencing phenomena associated with neuropathic pain: 'pins and needles' (paraesthesia); 'red skin' (autonomic changes); 'sensitive skin' (evoked dysaesthesia); 'electric shock pain'; and 'burning pain' (spontaneous dysaesthesia). The physical assessment (maximum score 8) is designed to identify allodynia by stroking cotton wool over the painful and the anatomically equivalent non-painful area, and altered pinprick threshold (PPT) by use of a 23-gauge needle to assess perception of pinprick in the same areas.

Adapted: The LANSS Pain Scale

f. PainDETECT Questionnaire

Date: **Patient:**

How would you assess your **pain** now, at this moment?

none max.

How strong was the **strongest** pain during the past 4 weeks?

0 1 2 3 4 5 6 7 8 9 10

none max.

How strong was the pain during the past 4 weeks on **average**?

0 1 2 3 4 5 6 7 8 9 10

none max.

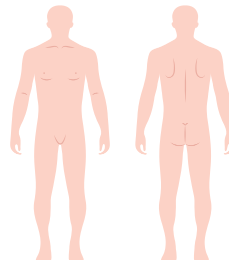
Mark the picture that best describes the course of your pain:

 Persistent pain with slight fluctuations ☐

 Persistent pain with pain attacks ☐

 Pain attacks without pain between them ☐

 Pain attacks with pain between them ☐



Does your pain radiate to other regions of your body?

Yes ☐ No ☐

If yes, please draw the direction in which the pain radiates.

Do you suffer from a burning sensation (e.g., stinging nettles) in the marked areas?
never ☐ hardly noticed ☐ slightly ☐ moderately ☐ strongly ☐ very strongly ☐

Do you have a tingling or prickling sensation in the areas of your pain (like crawling ants or electrical tingling)?
never ☐ hardly noticed ☐ slightly ☐ moderately ☐ strongly ☐ very strongly ☐

Is light touching (clothing, a blanket) in this area painful?
never ☐ hardly noticed ☐ slightly ☐ moderately ☐ strongly ☐ very strongly ☐

Do you have sudden pain attacks in the area of your pain, like electric shocks?
never ☐ hardly noticed ☐ slightly ☐ moderately ☐ strongly ☐ very strongly ☐

Is cold or heat (bath water) in this area occasionally painful?
never ☐ hardly noticed ☐ slightly ☐ moderately ☐ strongly ☐ very strongly ☐

Do you suffer from a sensation of numbness in the areas that you marked?
never ☐ hardly noticed ☐ slightly ☐ moderately ☐ strongly ☐ very strongly ☐

Does slight pressure in this area, e.g., with a finger, trigger pain?
never ☐ hardly noticed ☐ slightly ☐ moderately ☐ strongly ☐ very strongly ☐

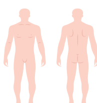
(To be filled out by the physician)

never	hardly notice	slightly	moderately	strongly	very strongly
x 0 =	x 1 =	x 2 =	x 3 =	x 4 =	x 5 =

Total Score ☐ **out of 35**

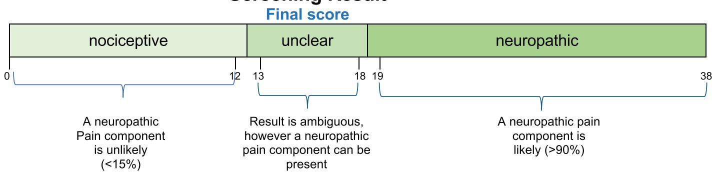
Please transfer the total score from the pain questionnaire:
Total Score [ ]

Please add up the total following numbers, depending on the marked pain behaviour and the pain radiation. Then total up the final score:

	Persistent pain with slight fluctuations	0	
	Persistent pain with pain attacks	-1	if marked, or
	Pain attacks without pain between them	+1	if marked, or
	Pain attacks with pain between them	+1	if marked
	Radiating pains?	+2	if yes

Final Score [ ]

Screening Result



The painDETECT questionnaire consists of seven questions that address the quality of neuropathic pain symptoms; it is completed by the patient and no physical examination is required. The first five questions ask about the gradation of pain, scored from 0 to 5 (never = 0, hardly noticed = 1, slightly = 2; moderately = 3, strongly = 4, very strongly = 5). Question 6 asks about the pain course pattern, scored from −1 to 2, depending on which pain course pattern diagram is selected. Question 7 asks about radiating pain, answered as yes or no, and scored as 2 or 0 respectively. The final score between −1 and 38, indicates the likelihood of a neuropathic pain component. A score of ≤ 12 indicates that pain is unlikely to have a neuropathic component (< 15%), while a score of ≥ 19 suggests that pain is likely to have a neuropathic component (> 90%).

Adapted: painDETECT Questionnaire.

g. Doeleur Neuropathique en 4 (DN4) Scale

To estimate the probability of neuropathic pain, please answer yes or no for each item of the following four questions.

INTERVIEW OF THE PATIENT

QUESTION 1:
Does the pain have one or more of the following characteristics?

	YES	**NO**
Burning……	☐	☐
Painful cold……	☐	☐
Electric shocks……	☐	☐

QUESTION 2:
Is the pain associated with one or more of the following symptoms in the same area?

	YES	**NO**
Tingling……	☐	☐
Pins and needles……	☐	☐
Numbness……	☐	☐
Itching……	☐	☐

EXAMINATION OF THE PATIENT

QUESTION 3:
Is the pain located in an area where the physical examination may reveal one or more of the following characterictics?

	YES	**NO**
Hypoesthesia to touch……	☐	☐
Hypoesthesia to pinprick……	☐	☐

QUESTION 4:
Is the painful area, can the pain be caused or increased by:

	YES	**NO**
Brushing?……	☐	☐

YES = 1 point
NO = 0 point

Patient's Score: /10

DN4 is a test for diagnosing neuropathic pain. It consists of 7 items related to symptoms and 3 related to clinical examination. The total DN4 score ranges from 0 to 10, and a score ≥ 4 indicates a diagnosis of peripheral neuropathic pain. The scores are added and a score of 4 or more out of 10 is suggestive of neuropathic pain.

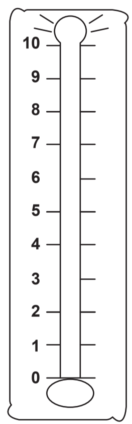
h. NCCN Distress Thermometer (English Version)
NCCN Guidelines Version 1.2024
Distress Management

NCCN DISTRESS THERMOMETER

Distress is an unpleasant experience of a mental, physical, social, or spiritual nature. It can affect the way you think, feel, or act. Distress may make it harder to cope with having cancer, its symptoms or its treatment.

Instructions: Please circle the number (0-10) that best describes how much distress you have been experiencing in the past week, including today.

Extreme distress



No distress

PROBLEM LIST

Have you had concerns about any of the items below in the past week, including today? (Mark all that apply)

<u>Physical Concerns</u>
- ❑ Pain
- ❑ Sleep
- ❑ Fatigue
- ❑ Tobacco use
- ❑ Substance use
- ❑ Memory or concentration
- ❑ Sexual health
- ❑ Changes in eating
- ❑ Loss or change of physical abilities

<u>Emotional Concerns</u>
- ❑ Worry or anxiety
- ❑ Sadnesss or depression
- ❑ Loss of interest or enjoyment
- ❑ Grief or loss
- ❑ Fear
- ❑ Loneliness
- ❑ Anger
- ❑ Changes in appearance
- ❑ Feelings of worthlessness or being a burden

<u>Social Concerns</u>
- ❑ Relationship with spouse or partner
- ❑ Relationship with children
- ❑ Relationship with family members
- ❑ Relationship with friends or coworkers
- ❑ Communication with healthcare team
- ❑ Ability to have children
- ❑ Prejudice or discrimination

<u>Practical Concerns</u>
- ❑ Taking care of myself
- ❑ Taking care of others
- ❑ Work
- ❑ School
- ❑ Housing
- ❑ Finances
- ❑ Insurance
- ❑ Transportation
- ❑ Child care
- ❑ Having enough food
- ❑ Access to medicine
- ❑ Treatment decisions

<u>Spiritual or Religious Concerns</u>
- ❑ Sense of meaning or purpose
- ❑ Changes in faith or beliefs
- ❑ Death, dying or afterlife
- ❑ Conflict between beliefs and cancer treatments
- ❑ Relationship with the sacred
- ❑ Ritual or dietary needs

<u>Other Concerns:</u>

____

____

____

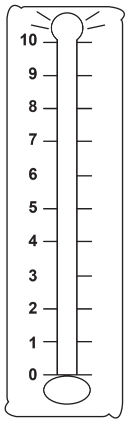
i. NCCN Distress Thermometer (Malay Version)
NCCN Guidelines Version 2.2022
Pengurusan Distres

TERMOMETER DISTRES NCCN

Distres merupakan satu pengalaman yang tidak menyenangkan, yang bersifat mental, fizikal, sosial atau spiritual. Distres boleh menjejaskan cara anda berfikir atau bertindak, atau perasaan anda. Distres boleh membuatkan seseorang berasa lebih sukar untuk menangani penyakit kanser, gejala-gejalanya, ataupun rawatannya.

Arahan: Sila bulatkan nombor (0-10) yang paling tepat menerangkan keadaan distress yang anda alami dalam tempoh seminggu yang lalu termasuk hari ini.

Distress keterlaluan



Tiada distress

SENARAI MASALAH

Adakah anda mempunyai kebimbangan tentang mana-mana item di bawah dalam tempoh seminggu yang lalu, termasuk hari ini? (Tandakan semua yang berkenaan)

Kebimbangan Fizikal
- ❑ Kesakitan
- ❑ Tidur
- ❑ Kelesuan
- ❑ Penggunaan tembakau
- ❑ Penggunaan bahan terlarang
- ❑ Daya ingatan atau penumpuan
- ❑ Kesihatan seksual
- ❑ Perubahan dalam pemakanan
- ❑ Kehilangan atau perubahan keupayaan fizikal

Kebimbangan Emosi
- ❑ Kerisauan atau keresahan
- ❑ Kesedihan atau kemurungan
- ❑ Kehilangan minat atau keseronokan
- ❑ Kehibaan atau kehilangan seseorang
- ❑ Ketakutan
- ❑ Kesunyian
- ❑ Kemarahan
- ❑ Perubahan dalam penampilan
- ❑ Rasa tidak berguna atau menyusahkan

Kebimbangan Sosial
- ❑ Hubungan dengan pasangan
- ❑ Hubungan dengan anak-anak
- ❑ Hubungan dengan ahli keluarga
- ❑ Hubungan dengan kawan-kawan atau rakan sekerja
- ❑ Hubungan dengan pasukan penjagaan kesihatan
- ❑ Komunikasi dengan pasukan penjagaan kesihatan
- ❑ Keupayaan mendapat anak

Kebimbangan Praktikal
- ❑ Menjaga diri sendiri
- ❑ Menjaga orang lain
- ❑ Pekerjaan
- ❑ Sekolah
- ❑ Perumahan
- ❑ Kewangan
- ❑ Insurans
- ❑ Pengangkutan
- ❑ Penjagaan anak
- ❑ Kecukupan makanan
- ❑ Akses kepada ubat-ubatan
- ❑ Keputusan rawatan

Kebimbangan Spiritual atau Keagamaan
- ❑ Maksud dan tujuan hidup
- ❑ Perubahan kepercayaan atau pegangan
- ❑ Kematian, akan mati atau selepas mati
- ❑ Konflik antara kepercayaan dan rawatan kanser
- ❑ Hubungan dengan Tuhan
- ❑ Ritual atau keperluan pemakanan

Kebimbangan Lain:

____

____

____

j. Integrated Palliative Care Outcome Scale (IPOS)

IPOS Patient Version

Name: ______________________________

Date (dd/mm/yyyy): ______________________________

Please write clearly, one letter or digit per box. Your answers will help us to keep improving your care and the care of others.

Thank you.

Q1. What have been your main problems or concerns over the past week?

1. ______________________________

2. ______________________________

3. ______________________________

Q2. Below is a list of symptoms, which you may or may not have experienced. For each symptom, please tick one box that best describes how it has affected you over the past week.

	Not at all	*Slightly*	*Moderately*	*Severely*	*Overwhelmingly*
Pain	0 ☐	1 ☐	2 ☐	3 ☐	4 ☐
Shortness of breath	0 ☐	1 ☐	2 ☐	3 ☐	4 ☐
Weakness or lack of energy	0 ☐	1 ☐	2 ☐	3 ☐	4 ☐
Nausea (feeling like you are going to be sick)	0 ☐	1 ☐	2 ☐	3 ☐	4 ☐
Vomiting (being sick)	0 ☐	1 ☐	2 ☐	3 ☐	4 ☐
Poor appetite	0 ☐	1 ☐	2 ☐	3 ☐	4 ☐
Constipation	0 ☐	1 ☐	2 ☐	3 ☐	4 ☐
Sore or dry mouth	0 ☐	1 ☐	2 ☐	3 ☐	4 ☐
Drowsiness	0 ☐	1 ☐	2 ☐	3 ☐	4 ☐
Poor mobility	0 ☐	1 ☐	2 ☐	3 ☐	4 ☐

Please list any other symptoms not mentioned above, and tick one box to show how they have affected you over the past week.

1. ____________	0 ☐	1 ☐	2 ☐	3 ☐	4 ☐
2. ____________	0 ☐	1 ☐	2 ☐	3 ☐	4 ☐
3. ____________	0 ☐	1 ☐	2 ☐	3 ☐	4 ☐

Over the past week:

	Not at all	*Occasionally*	*Sometimes*	*Most of the time*	*Always*
Q3. Have you been feeling anxious or worried about your illness or treatment?	0 ☐	1 ☐	2 ☐	3 ☐	4 ☐
Q4. Have any of your family or friends been anxious or worried about you?	0 ☐	1 ☐	2 ☐	3 ☐	4 ☐
Q5. Have you been feeling depressed?	0 ☐	1 ☐	2 ☐	3 ☐	4 ☐

	Always	*Most of the time*	*Sometimes*	*Occasionally*	*Not at all*
Q6. Have you felt at peace?	0 ☐	1 ☐	2 ☐	3 ☐	4 ☐
Q7. Have you been able to share how you are feeling with your family or friends as much as you wanted?	0 ☐	1 ☐	2 ☐	3 ☐	4 ☐
Q8. Have you had as much information as you wanted?	0 ☐	1 ☐	2 ☐	3 ☐	4 ☐

	Problems addressed/ No problems	*Problems mostly addressed*	*Problems partly addressed*	*Problems hardly addressed*	*Problems not addressed*
Q9. Have any practical problems resulting from your illness been addressed? (such as financial or personal)	0 ☐	1 ☐	2 ☐	3 ☐	4 ☐

	On my own	*With help from a a friend or relative*	*With help from a member or staff*
Q10. How did you complete this questionnaire?	☐	☐	☐

If you are worried about any of the issues raised on this questionnaire then please speak to your doctor or nurse.

IPOS (Integrated Palliative care Outcome Scale) is a measure of symptoms and concerns which matter to a patient. There are 10 questions scored on a scale of 1-4, which assess a patient's symptoms and needs with regards to physical, social, psychological and spiritual.

Adapted: IPOS Patient Version.

k. The Edmonton Symptom Assessment System (ESAS) Tool

Name:______________________________ **Phone Number:** _______________

Address:____________________________ **Completed By:**________________

Please circle a number that best describes how you feel:

Please circle a number that best describes how you feel:

0 1 2 3 4 5 6 7 8 9 10

No pain — **Worst possible pain**

0 1 2 3 4 5 6 7 8 9 10

Not tired — **Very tired**

0 1 2 3 4 5 6 7 8 9 10

No nausea — **Very nauseous**

0 1 2 3 4 5 6 7 8 9 10

Not depressed — **Very depressed**

0 1 2 3 4 5 6 7 8 9 10

Calm — **Very anxious**

0 1 2 3 4 5 6 7 8 9 10

Not drowsy — **Very drowsy**

0 1 2 3 4 5 6 7 8 9 10

Normal appetite — **No appetite**

0 1 2 3 4 5 6 7 8 9 10

Best feeling of well-being — **Worst possible feeling of well-being**

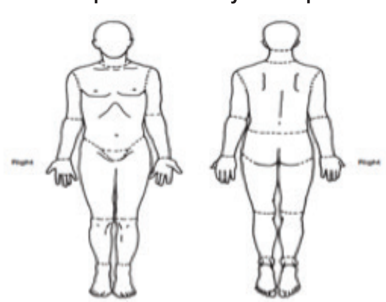
0 1 2 3 4 5 6 7 8 9 10

No shortness of breath — **Very short of breath**

0 1 2 3 4 5 6 7 8 9 10

Other problem

Please mark on these pictures where you feel pain or discomfort.



The ESAS is a comprehensive, yet brief and practical self-reporting tool of symptom severity (intensity) for nine common symptoms of advanced cancer (pain, tiredness, nausea, depression, anxiety, drowsiness, appetite, wellbeing, shortness of breath), with the option of adding a tenth patient-specific symptom. The time of assessment of each symptom is rated from 0 to 10 on a numerical scale, 0 meaning that the symptom is absent and 10 that it is of the worst possible severity.

Adapted: The Edmonton Symptom Assessment System (ESAS) Tool.

I. Pain Assessment in Advanced Dementia (PAINAD) Scale

Item*	0	1	2	Score
Breathing independent of vocalisation	Normal	Occasional laboured breathing. Short period of hyperventilation.	Noisy laboured breathing. Long period of hyperventilation. Cheyne-Stokes respirations.	
Negative vocalization	None	Occasional moan or groan. Low-level speech with a negative or disapproving quality.	Repeated troubled calling out. Loud moaning or groaning. Crying.	
Facial expression	Smiling or inexpressive	Sad. Frightened. Frown.	Facial grimacing.	
Body language	Relaxed	Tense. Distressed pacing. Fidgeting.	Rigid. Fists clenched. Knees pulled up. Pulling or pushing away. Striking out.	
Consolability	No need to console	Distracted or reassured by voice or touch.	Unable to console, distract or reassure.	
			Total**	

*Five-item observational tool (see the description of each item below).

**Total scores range from 0 to 10 (based on a scale of 0 to 2 for five items), with a higher score indicating more severe pain (0="no pain") to 10="severe pain")

The PAINAD scale is a reliable pain assessment tool for patients with advanced dementia. It assesses five behaviors: breathing, negative vocalisation, facial expression, body language, and the ability to be consoled. Each of the five indicators is scored on a range from 0 (not present) to 2 (completely present) based on direct observation for a total score that ranges from 0 to 10.

Reference: Warden V, Hurley AC, Volicer L. Development and psychometric evaluation of the Pain Assessment in Advanced Dementia (PAINAD) scale. J Am Med Dir Assoc. 2003;4(1):9-15.

Appendix 5

a. SUGGESTED MEDICATION DOSAGES AND ADVERSE EFFECTS IN ADULTS

Drug	Recommended Dosages	Side Effects	Remarks
Paracetamol	0.5 - 1 g, 6-8-hourly Max: 4 g/day	Rare	Consider dose reduction in hepatic impairment.
Non-Selective NSAIDs			
Diclofenac Sodium	50 - 150 mg daily, 8-12 -hourly Max: 150 mg/day	• Peptic ulcer • GI bleed • Platelet dysfunction • Renal impairment • Cardiac events	Use the lowest efficacious dose for the shortest possible duration. Consider dose reduction in renal impairment. Higher doses increase the risk of gastrointestinal and cardiovascular complications.
Mefenamic Acid	250 - 500 mg, 8-hourly Max: 1500 mg/day		
Ibuprofen	200 - 400 mg daily, 8-hourly Max: 2400 mg/day		
Selective NSAIDs			
Celecoxib	200 - 400 mg, 12 - 24-hourly Max: 400 mg/day	• Renal impairment • Cardiac events	Use the lowest efficacious dose for the shortest possible duration. Consider dose reduction in renal impairment and cardiovascular disease. Higher doses increase the risk of gastrointestinal and cardiovascular complications.
Etoricoxib	60 - 90 mg daily Max: 120 mg/day		
Weak Opioids			
Tramadol	50 - 100 mg, 6 - 8-hourly Max: 400 mg/day	• Drowsiness • Dizziness • Nausea • Vomiting • Constipation	Consider dose reduction in renal impairment.
Dihydrocodeine tartrate	30 - 60 mg, 6 - 8-hourly Max: 240 mg/day		
Combination Medications			
Paracetamol 500 mg + codeine 8 mg	1 - 2 tablets, 6 - 8-hourly Max: 8 tablets/day	• Drowsiness • Dizziness • Nausea • Vomiting • Constipation	Consider dose reduction in renal impairment and hepatic impairment.
Paracetamol 325 mg + tramadol 37.5 mg	1 - 2 tablets, 6 - 8-hourly Max: 8 tablets/day		
Strong Opioids			
Morphine	Starting dose (oral): 3 - 5 mg 4-hourly of IR morphine SR oral morphine: to be given in 12-hourly dosing	• Drowsiness • Dizziness • Nausea • Vomiting • Constipation	No max dose in cancer pain. Transdermal fentanyl can only be used when opioid requirements are stable, and never in an opioid naïve patient.
Oxycodone	Starting dose (oral): 5 mg of IR 4 - 6-hourly CR oxycodone: to be given in 12-hourly dosing		
Transdermal fentanyl	Equianalgesic dose of total 24 hours opioid requirement (refer to **Table 4** on conversion of opiods)		

Drug	Recommended Dosages	Side Effects	Remarks
Sublingual fentanyl	Starting dose: 100 mcg May be titrated up to 800 mcg/ episode of breakthrough pain For each episode of breakthrough pain, a second sublingual fentanyl tablet may be taken after 15-30 mins if there is inadequate pain relief (No further doses can be given. Inadequate pain relief after this dose would require other IR opioids). There is a lockout period of 2 hours before breakthrough pain can be treated again using sublingual fentanyl. Max: 2 doses/breakthrough pain episode; And 4 episodes of breakthrough pain within 24 hours	• Drowsiness • Dizziness • Nausea	The effectiveness and safety of doses above 800 mcg have not been evaluated. Oral mucositis or dry mouth may affect absorption.
Antidepressants			
Amitriptyline	Start with 12.5 - 25 mg ON Max: 150 mg/day	Anticholinergic effects e.g. dry mouth, drowsiness, urinary retention, arrhythmias, QT prolongation	Max dose seldom required. Usual effective dose 25 - 75 mg ON. Use with caution in the elderly and patients with cardiac disease, glaucoma, renal impairment and seizure risk.
Duloxetine	30 - 60 mg/day Max: 120 mg/day	• Bleeding risk • Hepatotoxicity (at higher doses) • Gastrointestinal disorder	Usual effective dose 60mg/day. Use with caution in patients with renal impairment and seizure risk.
Anticonvulsants			
Gabapentin	Start with 300 mg ON and increase by 300 mg/24 hrs every 2-3 days if necessary Max: 3600 mg/day	• Drowsiness, • dizziness, • amnesia, dry mouth tremor	Dose adjustment is required in renal impairment. Usual effective dose ≥ 600mg TDS
Pregabalin	50 - 150 mg BD Max dose: 300mg BD		Dose adjustment is required in renal impairment.
Bone Targeting Agents			
Zoledronic Acid	4 mg as a single IV infusion over 15 mins Can only be repeated after 7 days if response is inadequate	• Transient pyrexia & flu-like symptoms • Fatigue • Nausea • Osteonecrosis of the jaw	Consider dose reduction in renal impairment.

Drug	Recommended Dosages	Side Effects	Remarks
Pamidronate	30 - 90 mg as a single IV infusion over 2 - 4 hours Can only be repeated after 7 days if response is inadequate		
Denosumab	120 mg every 4 weeks	• Arthralgia • Fatigue • Hypocalcemia • Osteonecrosis of the jaw	
Corticosteroid			
Dexamethasone	8 - 16 mg/day (initial) Then reduce to lowest possible dose (usually 2mg/day)	• Bleeding • Susceptibility to infections • Impaired glycaemic control • Delirium & sleep disturbances	Try to give earlier in the day to minimise insomnia.
Laxatives			
Lactulose	15 - 20 ml orally, 6 - 8-hourly	• Bloating • Epigastric pain • Flatulence • Nausea • Vomiting • Cramping • Abdominal distension Nausea • Diarrhoea	May be mixed with fruit juice, water or milk. Reasonable fluid intake is required for effectiveness
Macrogol	1 - 4 sachets/day		Reasonable fluid intake is required for effectiveness
Bisacodyl	5 - 10 mg orally, 1 - 2 times daily Max: 20 mg/day	• Diarrhoea • Nausea • Vomiting • Rectal • irritation • Abdominal cramps • Bloating	Enteric coated tablet and should not be crushed. Exercise caution in GI obstruction, perforation or severe impaction.
Senna	2 tabs OD or 1 tab BD Max: 8 tabs/day	• Diarrhea • Nausea • Abdominal cramps	Exercise caution in GI obstruction, perforation or severe impaction.
Antiemetics			
Metoclopramide	10 - 20 mg, 6 - 8 hourly	• Extrapyramidal reactions • Dizziness • Drowsiness	Consider dose reduction in renal impairment.
Prochlorperazine	10 - 30 mg daily in divided doses		
Haloperidol	0.5 - 3 mg single dose nocte	• Extrapyramidal symptoms • Prolonged QT interval	

Adapted:

1. Wilcock A. Howard P, Charlesworth S. Palliative Care Formulary. Seventh Edition. London: Pharmaceutical Press; 2020.
2. Cherny NI, Fallon MT, Kaasa S, et al. Oxford Textbook of Palliative Medicine. Oxford: Oxford University Press; 2021.
3. Guidelines Review Committee. (2019). WHO Guidelines for the pharmacological and radiotherapeutic management of cancer pain in adults and adolescents. Available at: https://www.who.int/publications/i/item/9789241550390)

b. SUGGESTED MEDICATION DOSAGES IN PAEDIATRICS

Drug			Route	1 month-2 years	2 - 12 years	12 - 18 years
				Dose and frequency		
Non-opioids analgesics		Paracetamol	Oral	**0 - 3 months:** 15 mg/kg 6 - 8 H (Max: 60 mg/kg/day; if preterm 28 - 32 CGA, max 30 mg/kg/day) **>3 months** - 12 years: 15 mg/kg 4 - 6 H (Max: 75 mg/kg/day or 4 g/day)		500 mg - 1 g 4 - 6 g (if non-obese ≥50 kg: 1 g 4 -6H) (Max: 4 g/day)
Non-opioids analgesics		Paracetamol	Per rectal	**0 - 3 months: LD**: 30 mg/kg; **MD**: 20 mg/kg 8 H (Max: 60 mg/kg/day) **>3 month - 12 years: LD**: 40 mg/kg; **MD**: 15 - 20 mg/kg 6 H (Max: 75 mg/kg/day)		
Non-opioids analgesics		Paracetamol	IV	**Preterm neonate over 32/52 CGA**: 7.5 mg/kg 8 H (Max 25 mg/kg/day) **Term neonate & until 10 kg:** 7.5 mg/kg 6 - 8H (Max: 30 mg/kg/day) **>10 kg or child up to 50 kg:** 15 mg/kg 4 - 6H (Max: 60 mg/kg/day, not exceeding 2 g/day if <33 kg, or 3 g/day for 33 - 50 kg)		**If non-obese** >50 kg: 1 g 4 - 6H (Max: 4 g/day) ****Obese Children:** 15 mg/kg adjusted body weight (Max: 4 g/day)
Non-opioids analgesics	NSAIDs	Ibuprofen	Oral	**<3 months:** not recommended **>3 months:** 5 mg/kg 6 - 8 H (Max: 20 mg/kg/day) **6 months - 12 years**: 5 -10 mg/kg 6 - 8 H (Max: 30 - 40 mg/kg/day or 1.2 g/day, whichever is less)		200 mg - 400 mg 4 – 6 H (Max: 2.4 g/day)
Non-opioids analgesics	NSAIDs	Diclofenac	Oral	**<6 months:** not recommended **>6 months or >10 kg:** 0.3 - 1 mg/kg 8 H (Max: 3 mg/kg/day up to 150 mg/day, whichever is less for 2 days)		Oral 25 - 50 mg 8 H (Max: 3 doses/day)
Non-opioids analgesics	NSAIDs	Diclofenac	Per rectal	**> 1 year:** 1 mg/kg 8 - 12H (Max: 3 mg/kg/day up to 150 mg/day, whichever is less)		50 - 100 mg (oral to be started 18 H after initial 100 mg suppository)
Non-opioids analgesics	COX-2 inhibitors	Celecoxib	Oral	**<2 years:** not recommended	**>2 years:** weigh risks and benefits **10 - 25 kg:** 50 mg 12 H **>25 kg:** 100 mg 12 H	
Non-opioids analgesics	COX-2 inhibitors	Parecoxib	IV	**<2 years:** not recommended	**>2 years:** weigh risks and benefits A single dose of 0.5 -1 mg/kg (Max: 40 mg, administered by the anaesthetists in theatres. NSAIDs should only be administered 24 H after a dose of parecoxib.)	
		Fentanyl	IV	**For specialist use /with supervision only.** Bolus: 1 - 2 mcg/kg/dose (Should only be given by the Anaesthetists/PICU Team/Paediatric Emergency Specialist/ Trained accredited personnel) INFUSION in an independent line (in ICU): **Preparation:** Dilute 20 mcg/kg of fentanyl in 50 ml normal saline. 1 ml of solution= 0.4 mcg/kg of fentanyl Suggested rate: 0.5 - 2 ml/H (Max: 4 ml/H) (0.2 - 0.8 mcg/kg/H)		
		Fentanyl	PCA	(Restricted to Pain Service) **Initial PCA dosing:** Concentration: 0.4 mcg/ml; Bolus dose: 0.4 - 0.8 mcg/kg; Lockout interval: 5 to 7 minutes. Basal infusion (optional): 0 - 0.8 mcg/kg; 4 H limit - 4 mcg/kg		
		Tramadol	Oral/IV	**>1 year:** 0.5 - 1 mg/kg 4 - 6 H (with caution or refer Pain Service) **NB:** For tonsillectomy, max 1 mg/kg/dose 6 - 8 H (caution in OSA)		**>12 years:** 1 mg/kg 4 - 6 H (Max: 100 mg/dose or 400 mg/day)

	Drug	Route	1 month-2 years	2 - 12 years	12 - 18 years
			Dose and frequency		
	Oxycodone	Oral	**> 1 month:** IR 0.1 - 0.2 mg/kg (Max 5 mg) PRN or 4 - 6H (by APS/ Palliative team) **NB:** IR for acute pain; SR for severe background pain (only available in tablet)		
	Morphine	Oral	**> 1 month - 1 year:** 0.1 mg/kg 4 - 6 H (for moderate - severe pain) (use with caution) **> 1 year:** 0.1 - 0.2 mg/kg 4 - 6 H (for moderate pain) 0.2 - 0.4 mg/kg 4 - 6 H (for severe pain)		**>12 years:** 0.1 - 0.3 mg/kg 4 - 6 H (Max: 10 - 15mg/dose, up to 6 times/24 H)
	Morphine	SC	0.1- 0.2 mg/kg **<6 months:** (up to 4 times/24H) **>6 months:** (up to 6 times/24H)	0.2 mg/kg (up to 6 times/24 H)	**>12 years:** 5 - 10 mg (up to 6 times/24 H)
	Morphine	IV	**BOLUS:** **Slow titration: Refer IV morphine titration protocol** **1-12 months:** Max: 0.1 mg/kg (up to 4 times/24 H) **>1 year**: Max: 0.1 mg/kg (up to 6 times/ 24 H)		**BOLUS:** Slow titration: **Refer IV morphine titration protocol** **>12 years:** 2.5 - 10 mg (up to 6 times/24 H)
	Morphine	IV	**INFUSION in an independent line (with caution):** **Preparation**: Dilute 0.5 mg/kg of morphine (Max: 50 mg) in 50 ml normal saline. 1 ml of solution = 10 mcg/kg of morphine **Suggested rate:** **Neonates:** 0.5 - 0.7 ml/H (Max: 1 ml/H) (5 - 10 mcg/kg/H) **1 - 3 months:** 0.5 - 1 ml/H (Max: 2 ml/H) (5 - 20 mcg/kg/H) **>3 months:** 1 - 2ml/H (Max: 4ml/H) (10 - 40 mcg/kg/H)		
	Morphine	PCA	**Initial PCA dosing: (Restricted to APS team)** Concentration: 10 - 20 mcg/ml; Bolus dose: 10 - 20 mcg/kg; Lockout interval: 5 to 7 minutes. Basal infusion (optional): 0 - 20 mcg/kg/H; 4 H limit 300 mcg/kg		
	Naloxone	IV	0.01 mg/kg IV (Max: 0.4 mg) may repeat every 2 minutes		
Adjuvants	**Ketamine**	Oral	Oral: 2 - 10 mg/kg (sedation pre-medication)		
Adjuvants	**Ketamine**	IV	**BOLUS for analgesia:** 0.2 - 0.5 mg/kg; **for sedation:** 1 - 1.5 mg/kg (use restricted to trained personnel only) **INFUSION in an independent line:** (use restricted to trained personnel only) **Preparation:** Dilute 5 mg/kg of ketamine (Max: 250 mg) in 50 ml normal saline. 1 ml of solution= 100 mcg/kg of ketamine Suggested rate: 0.2 - 2 ml/H (Max: 4 ml/H) (20 - 400 mcg/kg/H)		
Adjuvants	**Clonidine**	Oral	**Analgesic adjunct:** 1 - 2 mcg/kg PRN or 8 H **sedation premedication:** 2 - 4 mcg/kg **NB:** antihypertensive - do not give if hypotensive		
Adjuvants	**Clonidine**	IV	1 - 2 mcg/kg PRN or 8H **(with caution)** **NB:** antihypertensive - do not give if hypotensive		
Adjuvants	**Gabapentin**	Oral	Initial dose: 5 mg/kg ON, increase if required to 5 mg/kg 12H (Day 2), then 5 mg/kg 8H (Day 3)		
Local Anaesthetics	**Lignocaine**	LA/RA	Max dose: 4 - 5mg/kg		
Local Anaesthetics	**Levobupivacaine/Bupivacaine**	LA/RA	Max dose: **Neonates - <6 months**: 1.5 - 2 mg/kg; **>6 months:** 2 - 2.5 mg/kg **NB:** bupivacaine is particularly cardiotoxic		
Local Anaesthetics	**Levobupivacaine/Bupivacaine**	Epidural infusion	**Levobupivacaine 0.1% ± fentanyl infusion:** (restricted to APS team) **Preparation:** Dilute 10 ml of levobupivacaine/bupivacaine 0.5% (i.e., 50 mg) in 50 ml normal saline + fentanyl **(<1 months**: Nil; **1 months - 1 year:** fentanyl 1 mcg/ml; **>1 year:** fentanyl 2 mcg/ml) Suggested rate: **Neonates:** levobupivacaine 0.1%: 0.1 - 0.2 ml/kg/H		

Drug		Route	1 month-2 years	2 - 12 years	12 - 18 years
			Dose and frequency		
			1 months - 1 year: levobupivacaine 0.1% + fentanyl 1 mcg/ml: 0.2 - 0.4 ml/kg/H **>1 year:** levobupivacaine 0.1% + fentanyl 2 mcg/ml: 0.2 - 0.4 ml/kg/H		
	Ropivacaine	LA/RA	Max dose: **Neonates** - **<6 months:** 1.5 - 2 mg/kg; **>6 months:** 2 - 3 mg/kg		
			Ropivacaine 0.1% ± fentanyl infusion: (restricted to APS team) Preparation: Dilute 25 ml of ropivacaine 0.2% (i.e., 50 mg) in 50 ml normal saline or dilute 6.7 ml of ropivacaine 0.75% (i.e., 50 mg) in 50 ml NS + fentanyl **(<1months:** Nil; **1 month - 1 year:** fentanyl 1 mcg/ml; **>1 year:** fentanyl 2 mcg/ml) **Suggested rate: Neonates:** ropivacaine 0.1%: 0.1 - 0.2 ml/kg/H **1 month - 1 year:** ropivacaine 0.1% + fentanyl 1 mcg/ml: 0.2 - 0.4 ml/kg/H **>1year:** ropivacaine 0.1% + fentanyl 2 mcg/ml: 0.2 - 0.4 ml/kg/H		

CGA: corrected gestational age; IR: immediate release; IV: intravenous; LA: local infiltration; NB: newborn; OSA: obstructive sleep apnoea; SC: subcutaneous; SR: slow release; PCA: patient-controlled analgesia; RA: regional anaesthesia; H: hour; Max: maximum; LD: loading dose; MD: maintenance dose; IBW: ideal body weight, ON: on night

****For Obese Children, recommended adjustments for drug dosing**:
Opioids: Ideal Body weight (IBW);
Paracetamol and NSAID: Adjusted Body Weight=IBW+ 0.4 x (Actual BW - IBW)

Adapted: Ministry of Health, Malaysia. Paediatric Pain Management Guidelines 2023. Putrajaya: MoH; 2023.

Appendix 6

GUIDE FOR TRANSDERMAL FENTANYL USE

Step 1: Preparation of the skin

- Ensure that the skin is completely dry and clean before applying the patch. Use only water to wash the skin. Do not apply soap, cream, oil or ointment on the area.
- Do NOT apply the patch over your Totally Implantable Venous Catheter (e.g. chemoport)/over a joint area/on irradiated skin.
- Body hair should be clipped with scissors IF necessary and NOT shaved.

Step 2: Preparation of patch

- Each patch is sealed in its own sachet. Tear or cut open the sachet at the notch/arrow. Do NOT cut across the middle of the sachet.

Step 3: Method of administration

- Peel one half of the plastic backing away from the centre of the patch. Try not to touch the sticky side of the patch. Press the sticky part of the patch onto the skin.
- Remove the other half of the plastic backing and press the whole patch onto the skin. Hold for 30 seconds. Make sure it sticks well, especially the edges.
- Change the patch every 72 hours. Remove the old patch before applying a new one. The date and time of patch due change should be written on the patch.
- Do NOT apply the patch on the same place twice in a row. Change the site of application to allow the skin to rest.
- If you need to apply more than one patch at a time, place the patches adequately apart so that the edges do not touch or overlap each other.
- If the patch falls off/peels off before the date and time of due change, apply a new patch on a new area of skin.

Step 3: Disposal of used patch

- Fold used patch in half with the adhesive side inwards.
- Discard in clinical waste bin (in the hospital) or in a waste bin at home and wash your hands.

Adapted: HKL Counseling Checklist (Oct 2020) and Hosp Selayang Palliative Unit Information Leaflet (Administration of Transdermal Fentanyl Patch).

Appendix 7

GUIDE FOR NALOXONE USE

GENERAL PRINCIPLES:
- Naloxone, a specific opioid antagonist, is seldom necessary in the palliative care setting when opioids are appropriately titrated against the patient's pain.
- It is indicated for the reversal of opioid-induced respiratory depression and not for treating drowsiness and/or delirium associated with opioids.
- The dose administered should be carefully titrated against level of consciousness and satisfactory respiratory function (≥8 breaths/ minute and no cyanosis).
- Titration is important to avoid acute withdrawal syndrome and severe pain.

PHARMACOKINETICS & AVAILABILITY:
- Route of administration: IV is preferable, but SC or IM can also be used
- Onset of action: 1 - 2 minutes (IV) and 2 - 5 minutes (SC/IM)
- Half-life: approximately 1 hour
- Pack size: 1 ampoule = 400 mcg/1 ml
- Adverse effects (usually with large bolus doses): abdominal cramps, nausea and vomiting, flushing, arrhythmias and erythema at injection site

TREATMENT:
- Respiratory depression is usually preceded by a progressive reduction in consciousness.
- If the respiratory rate ≥8 breaths/minute and patient can be easily aroused (e.g. opens eyes to verbal command), monitor patient closely and consider omitting or reducing the dose of the regular opioid.
- If respiratory rate is ≤8 breaths/min and patient is unresponsive, discontinue the ongoing opioid (e.g. stop CSCI/CIVI, remove TD patch) and naloxone should be administered.
    - o Dilute 1 ampoule (400 mcg) of naloxone in 10 ml water for injection
    - o Administer small boluses of 0.5 ml (20 mcg) every two minutes until respiratory rate is satisfactory, and patient is easily arousable (need not be fully alert)
    - o After the last dose of naloxone, continue to monitor the patient
    - o Further boluses of naloxone might be necessary because naloxone is shorter acting than the opioids.
    - o A naloxone infusion may be considered if recovery is not satisfactory with multiple bolus doses.

- After patient recovers, the regular opioid regimen must be reviewed to consider possible causes for the respiratory depression (e.g. drug interactions, drug accumulation due to renal impairment, medication errors) and necessary modifications made to the regimen.

ADDITIONAL CAUTIONS:
- Do not use large bolus doses e.g. “1 ampoule stat” in patients who are receiving opioids for chronic pain relief.
- Pupil size is an unreliable indicator of opioid overdose in patients taking regular opioids
- Naloxone should not be given to patients on opioids when death is expected and imminent; a slow respiratory rate is a normal occurrence.

Source: Wilcock A. Howard P, Charlesworth S. Palliative Care Formulary. Eighth Edition. London: Pharmaceutical Press; 2022.

LIST OF ABBREVIATIONS

AEs	adverse effects
ATC	around-the-clock
AUC	area under the curve
BD	two times a day
BPI	Brief Pain Inventory
BQ-II	The Barriers Questionnaire II
CAM	complementary and alternative medicine
CGA	corrected gestational age
CI	confidence interval
CPG	Clinical Practice Guidelines
CPN	coeliac plexus neurolysis
CNS	central nervous system
COMM	Current Opioid Misuse Measure
COX-2	cyclooxygenase-2
CR	controlled-release
CrCL	Creatinine Clearance
CV	cardiovascular
DN4	Doeleur Neuropathique en 4
EBRT	external beam therapy
eGFR	estimated glomerular filtration rate
EPCRC	European Palliative Care Research Collaborative
ESAS	Edmonton Symptom Assessment System
ESAS-CS	Edmonton Symptom Assessment System with additional symptoms of constipation, sleep and added time window of “past 24 hours”
ESAS-r-CS	Edmonton Symptom Assessment System with a time window of “now”
FLACC	Face Legs Activity Cry Consolability
FPS	Faces Pain Scale
FST	fentanyl sublingual tablets
g	gramme
GI	gastrointestinal
GRADE	Grading Recommendations, Assessment, Development and Evaluation
Gy	Gray (unit of ionising radiation dose)
IASP	International Association for the Study of Pain
IPOS	Integrated Palliative Care Outcome Scale
IR	immediate release
IV	intravenous
kg	kilogramme
LANSS	The Leeds Assessment of Neuropathic Symptoms and Signs
max	maximum
mcg	microgramme
MD	mean difference
MEDD	morphine milligramme equivalent daily dose
mg	milligramme
min	minute
ml	milliliter
MPQ	McGill Pain Questionnaire
MSAS	Memorial Symptom Assessment Scale
M3G	morphine-3-glucuronide
M6G	morphine-6-glucuronide

NCCN	National Comprehensive Cancer Network
NMDA	N-methyl-D-aspartate
NNH	number needed to harm
NNT	number needed to treat
NRS	Numeric Rating Scale
NRS-11	Numeric Rating Scale (0 - 10)
NSAIDs	nonsteroidal anti-inflammatory drugs
OD	daily
OMED	oral morphine equivalent daily dose
ONJ	osteonecrosis of the jaw
OR	odds ratio
ORT	Opioid Risk Tool
OTFC	oral transmucosal fentanyl citrate
PAINAD	Pain Assessment in Advanced Dementia tool
PFE	pain flare-effect
PO	by mouth
QoL	quality of life
RANK	receptor activator of nuclear factor kappa beta
RCT	randomised controlled trial
RD	risk difference
RR	risk ratio
SAEs	severe adverse events
SBRT	stereotactic body radiotherapy
SC	subcutaneous
SE	side effects
SHG	superior hypogastric plexus
SNRIs	selective norepinephrine reuptake inhibitors
SOAPP-R	Screener and Opioid Assessment for Patients with Pain-Revised
SR	sustained release
SRE	skeletal-related events
SMD	standardised mean difference
TD	transdermal
TDS	three times a day
TENS	Transcutaneous Electrical Nerve Stimulation
Tmax	time to peak drug concentration
VAS	Visual Analogue Scale
VRS	Verbal Rating Scale
WHO	World Health Organisation
WMD	Weighted mean difference
153Sm	(153Sm) lexidronam
186Re	rhenium (186Re) obisbemeda
188Re	Rhenium-188
223Ra	Radium-223

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